Definitions and Measurements of Polarization Mode Dispersion: Interferometric Versus Fixed Analyzer Methods

N. Gisin, R. Passy, and J. P. Von der Weid

***Abstract*—We compare, from a theoretical and an experimental point of view, the interferometric and the fixed analyzer (also known as the wavelength scanning) techniques for measuring polarization mode dispersion. The information provided by both techniques is shown to be identical, up to a Fourier transform. This information is related to a natural definition of polarization mode delay, $\Delta\tau$. For standard communication fibers, $\Delta\tau$ is itself related to the mean delay between the principal polarization modes, $\langle\tau(\omega)\rangle_\omega$, by a simple numerical factor.**

I. Introduction

The interest in Polarization Mode Dispersion (PMD) in single mode optical communication fibers is increasing, both because of the practical limitations due to severe constrains on linearity for CATV systems [1], and because of the long distance digital transmission cables. This interest is manifested in the activities of various standardization bodies (ITU, IEC, TIA [2]) to recommend definitions and measurement techniques for PMD. In this paper, we address both of these points. First we propose two definitions of PMD and recall the connection between them. Next we compare, theoretically and experimentally, the recently proposed fixed analyzer method [3], [4], also known as the wavelength scanning method, with the well-established interferometric technique. Essentially we show that the former carries out numerically the Fourier analysis that is done in the Fourier transform spectroscopy interferometer of the latter [5]. Finally, the definitions are related to the measurements data.

II. Definitions

The first definition of PMD, or rather of an appropriate average of Polarization Mode (PM) delay, closely follows the definition of modal dispersion for multimode fibers [6]. This is natural since the PM of a "single" mode fiber actually supports several modes—the polarization modes. In the absence of PM coupling, one has only two modes. But, in the more realistic case of random PM coupling, the fiber behaves as a many-mode fiber, and the natural definition for PM delay is the mean square deviation of the time of flight of the various PM. The difference here is that modal dispersion in multimode fibers is much larger than PMD in single mode fibers. In particular, the latter can easily be smaller than the coherence time of the light propagating down the fiber. Hence, in the definition of PMD, two main cases are to be considered according to the comparison between these two times. First we consider a definition neglecting the coherence of the light. Let $I(t)$ denote the time-dependent intensity at the output of the fiber when a short (with respect to the PM delay of the fiber) pulse is launched into the fiber. Then the PM delay is defined as twice the mean square deviation of this distribution (i.e., the full width)

$$PMDelay_1 = \Delta\tau = 2\left(\frac{\int I(t)t^2\,dt}{\int I(t)\,dt} - \left(\frac{\int I(t)t\,dt}{\int I(t)\,dt}\right)^2\right)^{1/2}. \tag{1}$$

For polarization maintaining fibers, as well as for many components, $I(t)$ has two peaks and $\Delta\tau$ is the delay between these peaks. In the case of high PM coupling, $I(t)$ is close to a Gaussian and $\Delta\tau$ is its full-width. For arbitrary random PM coupling, the general form of $I(t)$ can be computed [7]–[9] and $\Delta\tau$ expressed in terms of the fiber length l, the mean coupling length h, and the average modal birefringence D

$$(\Delta\tau)^2 = \frac{D^2h^2}{2}(2l/h - 1 + e^{-2l/h}). \tag{2}$$

For optical communication applications, the coherence time of the laser sources is, in general, larger than the PM delay. The outcoming PM modes will thus interfere. It can be shown that this interference pattern produces two outcoming pulses; each of them is polarized and their polarization states are called the principal PM [10], [11]. It is thus natural to consider, in addition to the distribution $I(t)$, the differential group delay $\tau(\omega)$ between these two modes. As is implicit in the notations, this differential group delay $\tau(\omega)$ is wavelength dependent. Accordingly, we propose a second definition for the PM delay based on the mean of $\tau(\omega)$ over a suitably large range of optical frequencies

$$PMDelay_2 = \langle\tau(\omega)\rangle_\omega = \frac{\int_{\omega_1}^{\omega_2}\tau(\omega)\,d\omega}{\omega_2 - \omega_1}. \tag{3}$$

The relations between these two definitions has been investigated theoretically [9] and experimentally [12]–[14]. The main

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result is that the differential group delays between the principal states $\tau(\omega)$ are related to our first definition as follows:

$$PMDelay_1 = \Delta\tau = \left(\frac{\int_{\omega_1}^{\omega_2} \tau(\omega)^2\, d\omega}{\omega_2 - \omega_1}\right)^{1/2}. \quad (4)$$

For polarization maintaining fibers, as well as for many components, the principal PM identifies with the usual eigenmodes, τ is thus essentially independent of the wavelength, and our two definitions coincide. For such fibers, the delay increases linearly with the fiber length [see (2) with $h > l$]. Accordingly,

$$PMD_{pm} = \frac{\Delta\tau}{l} = \frac{\langle\tau(\omega)\rangle_\omega}{l}\ \text{[ps/m]}. \quad (5)$$

Standard telecommunication fibers, on the contrary, have high PM couplings. In that case, the values of τ follow a Maxwell distribution [15]–[17] parameterized by $\Delta\tau$. Hence, using relation (4), our two definitions relate as follows: $\Delta\tau = 1.085\langle\tau(\omega)\rangle_\omega$, where the $1.085 = \sqrt{3\pi/8}$ numerical factor is the ratio between the mean of τ^2 and the mean of τ for τ distributed according to a Maxwell curve [14], [18]. For kilometer-long fibers, $l \gg h$, the delays increase only with the square root of the fiber length. Accordingly,

$$PMD_{l\gg h} = \frac{\Delta\tau}{\sqrt{l}} = 1.085\frac{\langle\tau(\omega)\rangle_\omega}{\sqrt{l}}[ps/\sqrt{\text{km}}]. \quad (6)$$

III. Comparison Between the Fixed Analyzer and Interferometric Measurement Techniques

We now come to the comparison between the fixed analyzer measurement technique and the interferometric one. Both can be implemented with a broadband light source followed by a polarizer before the fiber under test. In the interferometric case, the outcoming light is coupled to a Michelson interferometer, and the envelope $\mathcal{E}(t)$ of the fringe pattern is collected [8], [19]. As is well known from Fourier spectroscopy [5], this envelope is the Fourier transform of the outcoming light intensity $P(\omega)$. In the fixed analyzer technique, the light emerging from the fiber under test is passed through a second polarizer and analyzed with an Optical Spectrum Analyzer (OSA) [3], [4]. The detected signal $I(\omega)$ is then digitally Fourier transformed. Accordingly, one expects this Fourier transform $\mathcal{F}P$ to be the same as the fringe envelope $\mathcal{E}(t)$ obtained with the interferometric technique. This envelope $\mathcal{E}(t)$ is also the autocorrelation function of the intensity distribution $I(t)$ with a random amplitude $A(t)$ due to the random phases [8]

$$\mathcal{F}[P(\omega)](t) \approx \mathcal{E}(t) \approx A(t)\int I(t)I(t+\tau)\, d\tau. \quad (7)$$

The $PMDelay_1$ can thus be deduced from $\mathcal{E}(t)$ for the two cases of main practical interest. Indeed, for polarization maintaining fibers, $I(t)$ is sharply peaked; and for standard fibers with high PM coupling $I(t)$ has a Gaussian shape. In both cases, the deconvolution is straightforward. In the general case the deconvolution is more delicate, but $PMDelay_1$ can still be deduced from $\mathcal{E}(t)$ since the standard deviation of an autocorrelation is twice the standard deviation of the original function $I(t)$.

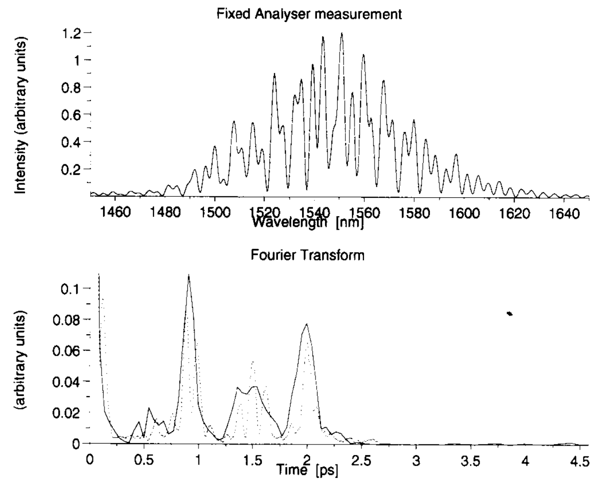

Fig. 1. Fixed analyzer measurement of two polarization maintaining fibers spliced together with an arbitrary angle. The spectral intensity $I(\omega)$ distribution as measured with a 0.1 nm resolution OSA is shown on the top, and its Fourier transform on the bottom. The dashed line is the fringe envelop $\mathcal{E}(t)$ obtained with the interferometric technique (actually the interferogram is symmetric around zero, but only half of it is shown).

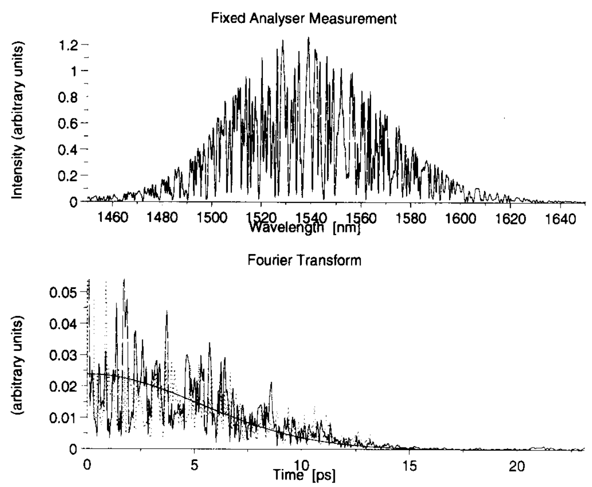

Fig. 2. Same as Fig. 1, but for a standard telecom optical cable. The Gaussian fit is also displayed.

Fig. 1 illustrates the relation between both techniques for the case of a concatenation of two polarization maintaining fibers spliced together with an arbitrary angle. Similarly, Fig. 2 shows the case of a standard telecom cable. The patterns show some fluctuations because we did not keep the input state of polarization fixed between the two measurements. But, anyway, the information about PMD is in the spread of the pattern, as defined in (1), and it agrees very well, within 0.1 ps, when using the same fitting procedures on the two sets of data.

A difficulty with the fixed analyzer method, common to all wavelength scanning methods, is that the spectral resolution

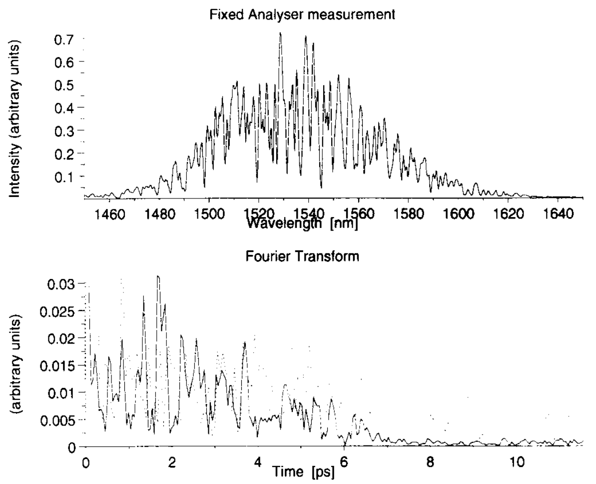

Fig. 3. Same as Fig. 2 for the same cable, but with a 1 nm spectral resolution. This illustrates that PM delay can be underestimated by using an insufficient resolution.

has to be small enough such that the corresponding coherence time is larger than the PM delay. Also, the number of points used to compute the Fourier transform has to be large enough to avoid any high frequencies cutoff. Fig. 3 illustrates the effect of insufficient spectral resolution. It concerns the same fiber as in Fig. 2, but with a tenfold wider spectral resolution. The result looks acceptable, but is actually wrong: it is only by controlling the resolution that one can avoid such underestimations of the PMD.

As for any polarimetric method, the fiber under test has to be motionless, thus excluding vibrations that happen, for instance, in ships during underwater cable installation or for aerial cables. This contrasts with the interferometric technique which is essentially nonsensitive to vibrations and for which the equivalent spectral resolution is determined by the scanning range. With our portable interferometer [20], for instance, we could measure PM delays up to more than 100 ps.

IV. Conclusion

We have shown that the interferometric and fixed analyzer measurement techniques provide the same information, either directly (interferometric) or via a digital Fourier transform (fixed analyzer). This information allows one to compute the PM delay as defined in (1). The interferometric technique shares the characteristics of Fourier spectroscopy [5], in particular a high dynamics since the detector collects all the light at each instance, and a large measurement range (up to 100 ps). A comparison between the interferometric technique and the Jones matrix analyses has already been presented in [14].

Under the realistic assumption of high PM coupling, as is the case for telecom cable, the average of the PM delay between the principal PM, i.e., our second definition ($PMDelay_2$) is related to our first definition ($PMDelay_1$) by a simple numerical factor.

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